LEGEND

- DUAL MOTOR FAN
- FAN WITH SINGLE MOTOR
- VOLUME CONTROL DAMPER
- MOTORIZED FIRE & SMOKE DAMPER TO UL555S
- MMD MOTORIZED MODULATING DAMPER
- BPRD BAROMETRIC PRESSURE RELIEF DAMPER
- HORIZONTAL AIR GRILLE
- VERTICAL AIR GRILLE/ WEATHERPROOF LOUVRE
- FIRE RATED ENCLOSURE (FRR 240/240/240) COMPLY WITH BS476:PART 24
- STAIRCASE/LOBBY PRESSURIZATION AIR DUCT F/B OR T/B
- STAIRCASE/LOBBY PRESSURIZATION AIR DUCT F/A OR T/A
- AIR RELEASE/PRESSURE RELIEF AIR DUCT F/B OR T/B
- AIR RELEASE/PRESSURE RELIEF AIR DUCT F/A OR T/A
- PS POINT TYPE SMOKE DETECTOR WITHIN 1M FROM THE DOOR
- SMOKE DETECTOR (PROBE TYPE)
- PS DIFFERENTIAL PRESSURE SENSOR TO BSEN 60654-1
- SCP SUPERVISORY CONTROL PANEL FOR STAIRCASE PRESSURIZATION SYSTEM
- AIR FLOW DIRECTION
- EXIT EXIT SIGN
- DP DIFFERENTIAL PRESSURE SWITCH
- LOCAL MOTOR CONTROL PANEL (LMCP) (C/W FIRE RATED ENCLOSURE (FRR 120/120/120) W/ SELF-CLOSING DOOR)
- NP NOTICE PLATE
- FLOOR MOUNTED WARNING SIGN (REFER TO DETAIL 1 OF 3/F LAYOUT- DWG NO. 2503373A/SPS/STB3/207)
- SPS AV SIGNAL FOR SPS (REFER TO DETAIL 2 OF 3/F LAYOUT- DWG NO. 2503373A/SPS/STB3/207)
- YELLOR & BLACK STRIPE CAUTION ADHESIVE FLOOR TAPE

B

ABBREVIATION:

ACC	ACCOMMODATION
AFA	AUTOMATIC FIRE ALARM
AFFL	ABOVE FINISHED FLOOR LEVEL
AP	ACCESS PANEL
ARD	AIR RELEASE DUCT
ARF	AIR RELEASE FAN (250°C, ONE HOUR FIRE RESISTANCE)
ARG	AIR RELEASE GRILLE
ARL	AIR RELEASE LOUVRE
C/W	COMPLETED WITH
DDC	DIRECT DIGITAL CONTROL
DN	DOWN
DWG	DRAWING
EA	EXHAUST AIR
EAD	EXHAUST AIR DUCT
FA	FRESH AIR
FAL	FRESH AIR INTAKE LOUVRE
FD	FIRE DAMPER
FR	FIRE RATED
FRP	FIRE RATED PROTECTION
FRR	FIRE RATED RESISTANCE
FRS	FIREFIGHTING & RESCUE STAIRWAY
FS	FIRE SERVICES
FSD	FIRE SERVICES DEPARTMENT
F/A	FROM ABOVE
F/B	FROM BELOW
H/L	HIGH LEVEL
LMCP	LOCAL MOTOR CONTORL PANEL
L/L	LOW LEVEL
MMD	MOTORIZED MODULATING DAMPER
MFSD	MOTORIZED FIRE AND SMOKE DAMPER
M/L	MID LEVEL
N/C	NORMAL CLOSE
N/O	NORMAL OPEN
PTSD	PROBE TYPE SMOKE DETECTOR
LPD	LOBBY PRESSURIZATION DUCT
LPF	LOBBY PRESSURIZATION FAN
LPG	LOBBY PRESSURIZATION GRILLE
LPL	LOBBY PRESSURIZATION LOUVRE
RA	RELEASE AIR
PRD	PRESSURE RELIEF DUCT
PRL	PRESSURE RELIEF LOUVRE
RP	RELIEF PRESSURE
SCP	SUPERVISORY CONTROL PANEL
SPD	STAIRCASE PRESSURIZTION DUCT
SPF	STAIRCASE PRESSURIZTATION FAN
SPG	STAIRCASE PRESSURIZTION GRILLE
SPL	STAIRCASE PRESSURIZTION LOUVRE
TYP.	TYPICAL
T/A	TO ABOVE
T/B	TO BELOW
TAD	TRANSFER AIR DUCT
VCD	VOLUME CONTROL DAMPER
V.D.	VENT DUCT
WP	WEATHERPROOF
B.L.	BOTTOM LEVEL
T.L.	TOP LEVEL
NP	NOTICE PLATE

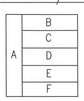
DESIGN CRITERIA

1. CLASS B SYSTEM, PROTECTION OF FIREFIGHTING SHAFT OF BS5588: PART 4, IS ADOPTED FOR THE DESIGN BASIS OF STAIRCASE AND LOBBY PRESSURIZATION SYSTEM.
2. INDEPENDENT PRESSURIZATION SYSTEM SHALL BE INSTALLED FOR THE REQUIRED STAIRCASE AND ITS ADJACENT FIREMAN LIFT LOBBIES. NO PRESSURIZATION SYSTEM WILL BE PROVIDED FOR THE CORRESPONDING FIREMAN' S LIFT SHAFT.
3. THE AIR SUPPLY SHALL BE SUFFICIENT TO MAINTAIN THE PRESSURE DIFFERENTIAL GIVEN BELOW WHEN ALL DOORS TO THE LIFT, STAIRCASE AND LOBBY AND THE FINAL EXIT DOOR ARE CLOSED AND THE AIR RELEASE PATH FROM THE ACCOMMODATION AREA IS OPEN AT THE FIRE FLOOR ONLY.

SPECIFIED AREA	PRESSURE DIFFERENTIAL TO BE MAINTAINED (MINIMUM)
ACROSS STAIRWAY AND ACCOMMODATION AREA	50 Pa
ACROSS CLOSED DOORS BETWEEN EACH FIREMAN'S LIFT LOBBY AND ACCOMMODATION AREA	45 Pa

4. CLASS B SYSTEM
THE AIR SUPPLY SHALL BE SUFFICIENT TO MAINTAIN AN AIRFLOW OF 2 M/S THROUGH THE OPEN DOORS BETWEEN THE LOBBY AND THE ACCOMMODATION AREA AT THE FIRE-AFFECTED STOREY WITH ALL THE FOLLOWING DOORS OPEN BETWEEN: -
   I. THE STAIR AND THE LOBBY ON THE FIRE-AFFECTED STOREY;
   II. THE STAIR AND THE LOBBY ON THE ADJACENT STOREY;
   III. THE FIREFIGHTING LIFT WELL AND THE LOBBY;
   IV. THE STAIR AND THE EXTERNAL AIR AT THE FIRE SERVICES ACCESS LEVEL;
   V. THE AIR RELEASE PATH FROM THE ACCOMMODATION AREA, ON THE STOREY ON WHICH THE AIR FLOW IS BEING MEASURED, I.E. FIRE-AFFECTED STOREY ONLY.
5. THE AIR FLOW VELOCITY SHALL BE MEASURED AT THE DOOR BETWEEN FIREMAN' S LIFT LOBBY AND ACCOMMODATION AREA ON FIRE-AFFECTED FLOOR.
6. THE MAXIMUM FORCE REQUIRED TO OPEN ANY DOOR WITHIN THE ESCAPE ROUTE SHALL BE IN NO CIRCUMSTANCES EXCEED 100 N IN ACCORDANCE WITH "CIRCULAR LETTER 2/2006".
7. FIRE COMPARTMENTATION SHALL BE ENSURED AT ALL TIMES.
8. ALL CONTROLS, STARTERS, RELAYS ETC., SHALL BE SUITABLE FOR CONTINUOUS OPERATION AT 250°C FOR NOT LESS THAN 1 HOUR. All ELECTRICAL POWER CABLES AND CONTROL CABLES USED SHALL BE FIRE RESISTANT CABLE. ALL CONTROL CABLES FOR MOTORIZED FIRE AND SMOKE DAMPERS SHALL BE FIRE RESISTANT CABLE.
9. PRESSURE SENSORS AND ASSOCIATED EQUIPMENT SHALL BE INDUSTRIAL PROCESS GRADE TO BSEN 60654-1.
10. ALL EQUIPMENT SERVING STAIRCASE PRESSURIZATION SYSTEM SHALL BE PROVIDED WITH AN ELECTRICAL SUPPLY FROM ESSENTIAL SOURCE.
11. SHEET METAL PRESSURIZATION AIR DUCT SHALL BE CONSTRUCTED UP TO A STANDARD OF HVCA DW144 OR SUBSEQUENT AMENDMENT AND AIR FLOW LEAKAGE TEST SHALL BE CONDUCTED UP TO A STANDARD OF HVCA DW 143 OR ITS AMENDMENT. FLANGE JOINT SHALL BE APPLIED TO SHEET METAL PRESSURIZATION AIR DUCTWORK WHICH SHALL CONFORM TO DW 144 REQUIREMENT.
12. WHENEVER FIRE RESISTING MATERIAL SUCH AS DURA STEEL OR PROMAT BOARD IS USED FOR PRESSURIZATION AIR DUCT CONSTRUCTION, THE THICKNESS AND CONSTRUCTION STANDARD SHALL FOLLOW THE MANUFACTURER' S RECOMMENDATION AND CONFORM TO BD AND FSD REQUIREMENTS.
13. THE PRIMARY AND SECONDARY POWER SUPPLIES WILL BE PROVIDED IN ACCORDANCE WITH "FSI-COP 2012(F1)
14. THE MOTORIZED FIRE & SMOKE DAMPER (MFSD) SHALL BE IN COMPLIANCE WITH UL555S.
15. FIRE RESISTANCE CABLES SHALL BE IN COMPLIANCE WITH THE STANDARD AS STATED IN FSD CIRCULAR LETTER 2/2017.
16. THE SWITCHBOARD SERVING FOR STAIRCASE PRESSURIZATION SYSTEM SHALL BE CONSTRUCTED IN ACCORDANCE WITH THE REQUIREMENT AS STIPULATED IN FSI-COP CLAUSE 5.21 F.3 (A) AND (B).
17. DEVICE FOR MONITORING OPEN/ CLOSE STATUS SHALL BE PROVIDED FOR ALL MFSD.
18. NO SMOKE DISCHARGE OR AIR RELEASE WITHIN 5M OF ANY DIRECTION OF AIR INTAKE OR OTHER OPENINGS INTO ANY BUILDING. NO DISCHARGES SHALL BE AT A HEIGHT ABOVE THE SURROUNDING HORIZONTAL SURFACE OF LESS THAN 3M TO THE BOTTOM OF THE OUTLET AND WHERE BELOW 6M SHALL NOT DISCHARGE DOWNWARDS.
19. SYSTEM DESIGN BASED ON CODE OF PRACTICE FOR MINIMUM FIRE SERVICES INSTALLATIONS AND EQUIPMENT - APRIL 2012,FSD CIRCULAR LETTER NO 2/2006 & 8/2021,& THE LATEST GBP SUBMISSION
B DATED 16 FEB 2026 (FSD REFERENCE: FP8 /31152 <122> DATED: 12 JUNE 2026).
20. FIRE RESISTANCE RATING OF ALL FIRE RATED ENCLOSURE /DUCTWORK SHALL COMPLY WITH FSI-COP 2012 AND COP FOR FIRE SAFETY IN BUILDINGS 2011.

GRILLE/DIFFUSER NOTATION



A = NUMBER
B = TYPE
- WP - WEATHERPROOF
- SPG - STAIRCASE PRESSURIZATION GRILLE
- LPG - LOBBY PRESSURIZATION GRILLE
- ARG - AIR RELEASE GRILLE
- SPL - STAIRCASE PRESSURIZATION LOUVER
- LPL - LOBBY PRESSURIZATION LOUVER
- ARL - AIR RELEASE LOUVER
- PRL - PRESSURE RELIEF LOUVER
- TAG - TRANSFER AIR GRILLE
- BPRD - BAROMETRIC PRESSURE RELIEF DAMPER
- *WEATHER PROOF LOUVRE EFFICIENCY IS ASSUMED TO BE 50% UNLESS OTHERWISE SPECIFIED.
- *AIR GRILLE EFFICIENCY IS ASSUMED TO BE 70%

C = DIMENSION (mm)
D = FLOW RATE (L/s)
E = FREE AREA VELOCITY (m/s)
F = BOTTOM LEVEL ABOVE FINISHED FLOOR LEVEL (mm)

COLOUR CODE

PRESSURE RELIEF DUCT	BLUE
LOBBY PRESSURIZATION DUCT	GREEN
STAIRCASE PRESSURIZTION DUCT	GREEN
AIR RELEASE DUCT	BROWN
TRANSFER AIR DUCT	ORANGE
FIRE RESISTING ENCLOSURE (FRR 240/240/240)	RED

FAN SCHEDULE FOR STAIRCASE PRESSURIZATION SYSTEM:

SERVED STAIRCASE	CLASS	FAN DESIGNATION	SERVED LEVEL AND AREA	FAN TYPE	FLOW RATE (L/S)	TEMPERATURE RATING
ST-B1	B	SPF-GF-01	B3-B1 STAIRCASE ST-B1	(SINGLE FAN DUAL MOTOR) FULLY BELTED	22900	-
		LPF-GF-01	B3-B1 FIREMAN'S LIFT LOBBY	(SINGLE FAN DUAL MOTOR) FULLY BELTED	3800	-
		ARF-GF-01A	B3-B1 ACCOMMODATION AREA (DUTY)	(SINGLE FAN SINGLE MOTOR)	10700	250 °C FOR AT LEAST 1 HOURS.
		ARF-GF-01B	B3-B1 ACCOMMODATION AREA (STANDBY)	(SINGLE FAN SINGLE MOTOR)	10700	

DRAWING LIST:

DRAWING NO.	REV.	DRAWING TITLE	SCALE
2503373A/SPS/STB1/001	B	STAIRCASE PRESSURIZATION SYSTEM - GENERAL NOTES, LEGEND, ABBREVIATION, DRAWING LIST, BLOCK PLAN AND FAN SCHEDULE	N.T.S.
2503373A/SPS/STB1/101	B	STAIRCASE PRESSURIZATION SYSTEM - SCHEMATIC DIAGRAM FOR ST-B1	N.T.S.
2503373A/SPS/STB1/102	A	STAIRCASE PRESSURIZATION SYSTEM - SUPERVISORY CONTROL PANEL FOR ST-B1	N.T.S.
2503373A/SPS/STB1/103	A	STAIRCASE PRESSURIZATION SYSTEM - FAN CONTROL PANEL FOR ST-B1	N.T.S.
2503373A/SPS/STB1/201	B	STAIRCASE PRESSURIZATION SYSTEM - B3 LAYOUT (ST-B1)	1:100@A0
2503373A/SPS/STB1/202	B	STAIRCASE PRESSURIZATION SYSTEM - B2 LAYOUT (ST-B1)	1:100@A0
2503373A/SPS/STB1/203	B	STAIRCASE PRESSURIZATION SYSTEM - B1 LAYOUT (ST-B1)	1:100@A0
2503373A/SPS/STB1/204	B	STAIRCASE PRESSURIZATION SYSTEM - GF LAYOUT (ST-B1)	1:100@A0
2503373A/SPS/STB1/205	0	STAIRCASE PRESSURIZATION SYSTEM - 1F LAYOUT (ST-B1)	1:100@A0
2503373A/SPS/STB1/206	0	STAIRCASE PRESSURIZATION SYSTEM - 2F LAYOUT (ST-B1)	1:100@A0
2503373A/SPS/STB1/207	A	STAIRCASE PRESSURIZATION SYSTEM - 3F LAYOUT (ST-B1)	1:100@A0
2503373A/SPS/STB1/301	B	STAIRCASE PRESSURIZATION SYSTEM - EAST ELEVATION PLAN	1:100@A0
2503373A/SPS/STB1/302	A	STAIRCASE PRESSURIZATION SYSTEM - WEST ELEVATION PLAN	1:100@A0
2503373A/SPS/STB1/401	A	STAIRCASE PRESSURIZATION SYSTEM - ELECTRICAL & LMCP WIRING DIAGRAM(SHEET 1 OF 4)(FOR REFERENCE ONLY)	N.T.S.
2503373A/SPS/STB1/402	B	STAIRCASE PRESSURIZATION SYSTEM - ELECTRICAL & LMCP WIRING DIAGRAM(SHEET 2 OF 4)(FOR REFERENCE ONLY)	N.T.S.
2503373A/SPS/STB1/403	B	STAIRCASE PRESSURIZATION SYSTEM - ELECTRICAL & LMCP WIRING DIAGRAM(SHEET 3 OF 4)(FOR REFERENCE ONLY)	N.T.S.
2503373A/SPS/STB1/404	A	STAIRCASE PRESSURIZATION SYSTEM - ELECTRICAL & LMCP WIRING DIAGRAM(SHEET 4 OF 4)(FOR REFERENCE ONLY)	N.T.S.

B

EQUIPMENT NOTATION



(D)
(E)

A = EQUIPMENT ABBREVIATION
- LPF - LOBBY PRESSURIZATION FAN
- SPF - STAIRCASE PRESSURIZATION FAN
- ARF - AIR RELEASE FAN

B = LEVEL (FLOOR)
C = UNIT NUMBER
D = FLOW RATE (L/s)
E = ADDITIONAL INFORMATION (IF ANY)

A-B-C-D

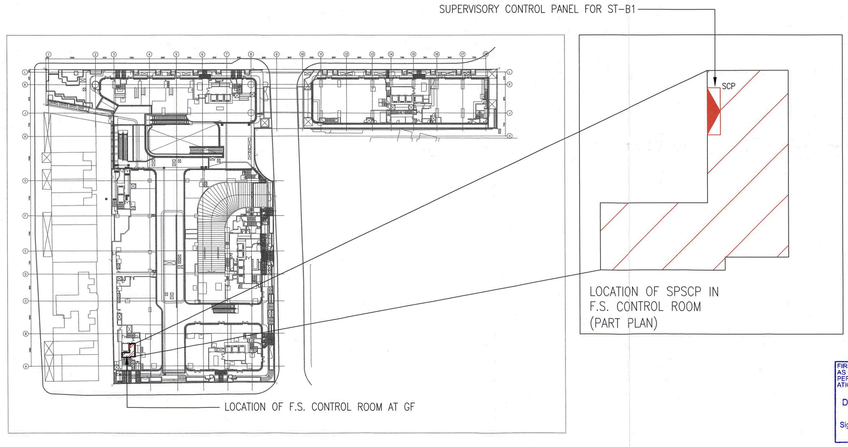
A = ABBREVIATION
- MFSD - MOTORIZED FIRE AND SMOKE DAMPER
- MMD - MOTORIZED MODULATING DAMPER
- PTSD - PROBE TYPE SMOKE DETECTOR

B = INSTALLATION POSITION
- LPF - LOBBY PRESSURIZATION FAN
- LPG - LOBBY PRESSURIZATION GRILLE
- SPF - STAIRCASE PRESSURIZATION FAN
- SPG - STAIRCASE PRESSURIZATION GRILLE
- ARF - AIR RELEASE FAN
- ARG - AIR RELEASE GRILLE

C = LEVEL (FLOOR)
D = UNIT NUMBER

Mtwl 18.8.26

LOO Kwan Wai
(RPE NO.: RP0260258)
18.8.26



BLOCK PLAN AT GROUND FLOOR(G/F) SCALE 1:2000

FIRE SERVICE INSTALLATION PLANS AS CERTIFIED BY THE AUTHORIZED PERSON AND FIRE SERVICE INSTALLATION CONTRACTOR / CONSULTANT
Date: 27 AUG 2026
Signed: (FUNG Kam-wah)
for Director of Fire Services

RETURN

Mtwl
LOO Kwan Wai
(RPE No.: RP0260258)

NO.	DATE	REVISIONS	DRAFTED	CHECKED	DESIGNED	REVIEWED
B	06/2026	FSD THIRD SUBMISSION	HYH	WWC	HYH	TKL
A	07/2025	FSD SECOND SUBMISSION	HYH	WWC	HYH	TKL
0	03/2025	FSD FIRST SUBMISSION	HYH	WWC	HYH	TKL

CLIENT:

NATION STAR DEVELOPMENT LIMITED
75/F, 8 Finance Street,
Central, Hong Kong

PROJECT ARCHITECT & AUTHORIZED PERSON:
RONALD LU & PARTNERS
呂元祥建築師事務所
33rd Floor, Wu Chung house, 213 Queen's Road East, Wanchai, Hong Kong

STRUCTURAL/GEOTECHNICAL
wsp
7/F, One Kowloon
1 Wang Yuen Street
Kowloon Bay, Hong Kong

BUILDING SERVICES ENGINEER:
wsp
7/F, One Kowloon
1 Wang Yuen Street
Kowloon Bay, Hong Kong

QUANTITY SURVEYOR:
ARCADIS
17/F, Two Harbour Square, 180 Wai Yip Street,
Kwun Tong, Kowloon
Hong Kong

FIRE SERVICES CONTRACTOR:
開創工程有限公司
GRAND CREATIVE ENGINEERS LIMITED
CLASS I, II, III REGISTERED FIRE SERVICES CONTRACTOR
WSD LICENSED PLUMBER I GRADE 1 2
EMSD REGISTERED ELECTRICAL CONTRACTOR
陳廣開
Chan Kwong Hoi
Director
Unit D, 2/F, Cheong Wah Comm. Bldg., 900-904 Canton Road, Kowloon.
Mobile : 9016 6066 Tel : 3512 5782 Fax : 3612 5785
Email : info@grandcreative.com.hk

PROJECT:
PROPOSED DEVELOPMENT AT BAILEY STREET & WING KWONG STREET, TO KWA WAN, KOWLOON
INLAND LOT NO. 11279
(URA PROJECT KC-009)

TITLE:
STAIRCASE PRESSURIZATION SYSTEM- GENERAL NOTES, LEGEND, ABBREVIATION, DRAWING LIST, BLOCK PLAN & FAN SCHEDULE

DATE: 06/2026	SCALE: NTS@A0		
DRAFTED: HYH	CHECKED: WWC	DESIGNED: HYH	REVIEWED: TKL
JOB NO. 2503373A	DRAWING NO. 2503373A/SPS/STB1/001		REV. NO. B